

# YOLO Classification ONNX Model Inference Report

Backend Engine: ONNX

## Phase 1: Configuration & System Metadata

|                         |                                 |
|-------------------------|---------------------------------|
| Model Name              | yolo26n-cls.onnx                |
| Input Dimensions        | 224x224                         |
| Label Map               | /workspace/cls-labels/label.txt |
| Source Images Directory | /workspace/Images               |
| Processed Images Output | /workspace/ONNX_Output          |
| OS                      | Ubuntu 24.04.4 LTS              |
| Python Version          | 3.12.3                          |
| PyTorch Version         | 2.13.0+cu130                    |
| Ultralytics Version     | 8.4.75                          |
| Inference Device        | NVIDIA RTX A4000                |
| CUDA Version            | 13.0                            |
| Total Parameters        | 2,807,042                       |
| File Size               | 10.82 MB                        |
| Number of Classes       | 1000                            |
| Precision Mode          | FP32                            |

### Model Trained Class Names:

tench, goldfish, great\_white\_shark, tiger\_shark, hammerhead, electric\_ray, stingray, cock, hen, ostrich, brambling, goldfinch, house\_finch, junco, indigo\_bunting, robin, bulbul, jay, magpie, chickadee, water\_ouzel, kite, bald\_eagle, vulture, great\_grey\_owl, European\_fire\_salamander, common\_newt, eft, spotted\_salamander, axolotl, bullfrog, tree\_frog, tailed\_frog, loggerhead, leatherback\_turtle, mud\_turtle, terrapin, box\_turtle, banded\_gecko, common\_iguana, American\_chameleon, whiptail, agama, frilled\_lizard, alligator\_lizard, Gila\_monster, green\_lizard, African\_chameleon, Komodo\_dragon, African\_crocodile, American\_alligator, triceratops, thunder\_snake, ringneck\_snake, hognose\_snake, green\_snake, king\_snake, garter\_snake, water\_snake, vine\_snake, night\_snake, boa\_constrictor, rock\_python, Indian\_cobra, green\_mamba, sea\_snake, horned\_viper, diamondback, sidewinder, trilobite, harvestman, scorpion, black\_and\_gold\_garden\_spider, barn\_spider, garden\_spider, black\_widow, tarantula, wolf\_spider, tick, centipede, black\_grouse, ptarmigan, ruffed\_grouse, prairie\_chicken, peacock, quail, partridge, African\_grey, macaw, sulphur-crested\_cockatoo, lorikeet, coucal, bee\_eater, hornbill, hummingbird, jacamar, toucan, drake, red-breasted\_merganser, goose, black\_swan, tusker, echidna, platypus, wallaby, koala, wombat, jellyfish, sea\_anemone, brain\_coral, flatworm, nematode, conch, snail, slug, sea\_slug, chiton, chambered\_nautilus, Dungeness\_crab, rock\_crab, fiddler\_crab, king\_crab, American\_lobster, spiny\_lobster, crayfish, hermit\_crab, isopod, white\_stork, black\_stork, spoonbill, flamingo, little\_blue\_heron, American\_egret, bittern, crane\_(bird), limpkin, European\_gallinule, American\_coot, bustard, ruddy\_turnstone, red-backed\_sandpiper, redshank, dowitcher,

oystercatcher, pelican, king\_penguin, albatross, grey\_whale, killer\_whale, dugong, sea\_lion, Chihuahua, Japanese\_spaniel, Maltese\_dog, Pekinese, Shih-Tzu, Blenheim\_spaniel, papillon, toy\_terrier, Rhodesian\_ridgeback, Afghan\_hound, basset, beagle, bloodhound, bluetick, black-and-tan\_coonhound, Walker\_hound, English\_foxhound, redbone, borzoi, Irish\_wolfhound, Italian\_greyhound, whippet, Ibizan\_hound, Norwegian\_elkhound, otterhound, Saluki, Scottish\_deerhound, Weimaraner, Staffordshire\_bullterrier, American\_Staffordshire\_terrier, Bedlington\_terrier, Border\_terrier, Kerry\_blue\_terrier, Irish\_terrier, Norfolk\_terrier, Norwich\_terrier, Yorkshire\_terrier, wire-haired\_fox\_terrier, Lakeland\_terrier, Sealyham\_terrier, Airedale, cairn, Australian\_terrier, Dandie\_Dinmont, Boston\_bull, miniature\_schnauzer, giant\_schnauzer, standard\_schnauzer, Scotch\_terrier, Tibetan\_terrier, silky\_terrier, soft-coated\_wheaten\_terrier, West\_Highland\_white\_terrier, Lhasa, flat-coated\_retriever, curly-coated\_retriever, golden\_retriever, Labrador\_retriever, Chesapeake\_Bay\_retriever, German\_short-haired\_pointer, vizsla, English\_setter, Irish\_setter, Gordon\_setter, Brittany\_spaniel, clumber, English\_springer, Welsh\_springer\_spaniel, cocker\_spaniel, Sussex\_spaniel, Irish\_water\_spaniel, kuvasz, schipperke, groenendael, malinois, briard, kelpie, komondor, Old\_English\_sheepdog, Shetland\_sheepdog, collie, Border\_collie, Bouvier\_des\_Flandres, Rottweiler, German\_shepherd, Doberman, miniature\_pinscher, Greater\_Swiss\_Mountain\_dog, Bernese\_mountain\_dog, Appenzeller, EntleBucher, boxer, bull\_mastiff, Tibetan\_mastiff, French\_bulldog, Great\_Dane, Saint\_Bernard, Eskimo\_dog, malamute, Siberian\_husky, dalmatian, affenpinscher, basenji, pug, Leonberg, Newfoundland, Great\_Pyrenees, Samoyed, Pomeranian, chow, keeshond, Brabancon\_griffon, Pembroke, Cardigan, toy\_poodle, miniature\_poodle, standard\_poodle, Mexican\_hairless, timber\_wolf, white\_wolf, red\_wolf, coyote, dingo, dhole, African\_hunting\_dog, hyena, red\_fox, kit\_fox, Arctic\_fox, grey\_fox, tabby, tiger\_cat, Persian\_cat, Siamese\_cat, Egyptian\_cat, cougar, lynx, leopard, snow\_leopard, jaguar, lion, tiger, cheetah, brown\_bear, American\_black\_bear, ice\_bear, sloth\_bear, mongoose, meerkat, tiger\_beetle, ladybug, ground\_beetle, long-horned\_beetle, leaf\_beetle, dung\_beetle, rhinoceros\_beetle, weevil, fly, bee, ant, grasshopper, cricket, walking\_stick, cockroach, mantis, cicada, leafhopper, lacewing, dragonfly, damselfly, admiral, ringlet, monarch, cabbage\_butterfly, sulphur\_butterfly, lycaenid, starfish, sea\_urchin, sea\_cucumber, wood\_rabbit, hare, Angora, hamster, porcupine, fox\_squirrel, marmot, beaver, guinea\_pig, sorrel, zebra, hog, wild\_boar, warthog, hippopotamus, ox, water\_buffalo, bison, ram, bighorn, ibex, hartebeest, impala, gazelle, Arabian\_camel, llama, weasel, mink, polecat, black-footed\_ferret, otter, skunk, badger, armadillo, three-toed\_sloth, orangutan, gorilla, chimpanzee, gibbon, siamang, guenon, patas, baboon, macaque, langur, colobus, proboscis\_monkey, marmoset, capuchin, howler\_monkey, titi, spider\_monkey, squirrel\_monkey, Madagascar\_cat, indri, Indian\_elephant, African\_elephant, lesser\_panda, giant\_panda, barracouta, eel, coho, rock\_beauty, anemone\_fish, sturgeon, gar, lionfish, puffer, abacus, abaya, academic\_gown, accordion, acoustic\_guitar, aircraft\_carrier, airliner, airship, altar, ambulance, amphibian, analog\_clock, apiary, apron, ashcan, assault\_rifle, backpack, bakery, balance\_beam, balloon, ballpoint, Band\_Aid, banjo, bannister, barbell, barber\_chair, barbershop, barn, barometer, barrel, barrow, baseball, basketball, bassinet, bassoon, bathing\_cap, bath\_towel, bathtub, beach\_wagon, beacon, beaker, bearskin, beer\_bottle, beer\_glass, bell\_cote, bib, bicycle-built-for-two, bikini, binder, binoculars, birdhouse, boathouse, bobsled, bolo\_tie, bonnet, bookcase, bookshop, bottlecap, bow, bow\_tie, brass, brassiere, breakwater, breastplate, broom, bucket, buckle, bulletproof\_vest, bullet\_train, butcher\_shop, cab, caldron, candle, cannon, canoe, can\_opener, cardigan, car\_mirror, carousel, carpenter's\_kit, carton, car\_wheel, cash\_machine, cassette, cassette\_player, castle, catamaran, CD\_player, cello, cellular\_telephone, chain, chainlink\_fence, chain\_mail, chain\_saw, chest, chiffonier, chime, china\_cabinet, Christmas\_stocking, church, cinema, cleaver, cliff\_dwelling, cloak, clog, cocktail\_shaker, coffee\_mug, coffeepot, coil, combination\_lock, computer\_keyboard, confectionery, container\_ship, convertible, corkscrew, cornet, cowboy\_boot, cowboy\_hat, cradle, crane\_(machine), crash\_helmet, crate, crib, Crock\_Pot, croquet\_ball, crutch, cuirass, dam, desk, desktop\_computer, dial\_telephone, diaper, digital\_clock, digital\_watch, dining\_table, dishrag, dishwasher, disk\_brake, dock, dogsled, dome, doormat, drilling\_platform, drum, drumstick, dumbbell, Dutch\_oven, electric\_fan, electric\_guitar, electric\_locomotive, entertainment\_center, envelope, espresso\_maker, face\_powder, feather\_boa, file, fireboat, fire\_engine, fire\_screen, flagpole, flute, folding\_chair, football\_helmet, forklift, fountain, fountain\_pen, four-poster, freight\_car, French\_horn, frying\_pan, fur\_coat, garbage\_truck, gasmask, gas\_pump, goblet, go-kart, golf\_ball, golfcart, gondola, gong, gown, grand\_piano, greenhouse, grille, grocery\_store, guillotine, hair\_slide, hair\_spray, half\_track, hammer, hamper, hand\_blower, hand-held\_computer, handkerchief, hard\_disc, harmonica, harp, harvester, hatchet, holster, home\_theater,

honeycomb, hook, hoopskirt, horizontal\_bar, horse\_cart, hourglass, iPod, iron, jack-o'-lantern, jean, jeep, jersey, jigsaw\_puzzle, jinrikisha, joystick, kimono, knee\_pad, knot, lab\_coat, ladle, lampshade, laptop, lawn\_mower, lens\_cap, letter\_opener, library, lifeboat, lighter, limousine, liner, lipstick, Loafer, lotion, loudspeaker, loupe, lumbermill, magnetic\_compass, mailbag, mailbox, maillot\_(tights), maillot\_(tank\_suit), manhole\_cover, maraca, marimba, mask, matchstick, maypole, maze, measuring\_cup, medicine\_chest, megalith, microphone, microwave, military\_uniform, milk\_can, minibus, miniskirt, minivan, missile, mitten, mixing\_bowl, mobile\_home, Model\_T, modem, monastery, monitor, moped, mortar, mortarboard, mosque, mosquito\_net, motor\_scooter, mountain\_bike, mountain\_tent, mouse, mousetrap, moving\_van, muzzle, nail, neck\_brace, necklace, nipple, notebook, obelisk, oboe, ocarina, odometer, oil\_filter, organ, oscilloscope, overskirt, oxcart, oxygen\_mask, packet, paddle, paddlewheel, padlock, paintbrush, pajama, palace, panpipe, paper\_towel, parachute, parallel\_bars, park\_bench, parking\_meter, passenger\_car, patio, pay-phone, pedestal, pencil\_box, pencil\_sharpener, perfume, Petri\_dish, photocopier, pick, pickelhaube, picket\_fence, pickup, pier, piggy\_bank, pill\_bottle, pillow, ping-pong\_ball, pinwheel, pirate, pitcher, plane, planetarium, plastic\_bag, plate Rack, plow, plunger, Polaroid\_camera, pole, police\_van, poncho, pool\_table, pop\_bottle, pot, potter's\_wheel, power\_drill, prayer\_rug, printer, prison, projectile, projector, puck, punching\_bag, purse, quill, quilt, racer, racket, radiator, radio, radio\_telescope, rain\_barrel, recreational\_vehicle, reel, reflex\_camera, refrigerator, remote\_control, restaurant, revolver, rifle, rocking\_chair, rotisserie, rubber\_eraser, rugby\_ball, rule, running\_shoe, safe, safety\_pin, saltshaker, sandal, sarong, sax, scabbard, scale, school\_bus, schooner, scoreboard, screen, screw, screwdriver, seat\_belt, sewing\_machine, shield, shoe\_shop, shoji, shopping\_basket, shopping\_cart, shovel, shower\_cap, shower\_curtain, ski, ski\_mask, sleeping\_bag, slide\_rule, sliding\_door, slot, snorkel, snowmobile, snowplow, soap\_dispenser, soccer\_ball, sock, solar\_dish, sombrero, soup\_bowl, space\_bar, space\_heater, space\_shuttle, spatula, speedboat, spider\_web, spindle, sports\_car, spotlight, stage, steam\_locomotive, steel\_arch\_bridge, steel\_drum, stethoscope, stole, stone\_wall, stopwatch, stove, strainer, streetcar, stretcher, studio\_couch, stupa, submarine, suit, sundial, sunglass, sunglasses, sunscreen, suspension\_bridge, swab, sweatshirt, swimming\_trunks, swing, switch, syringe, table\_lamp, tank, tape\_player, teapot, teddy, television, tennis\_ball, thatch, theater\_curtain, thimble, thresher, throne, tile\_roof, toaster, tobacco\_shop, toilet\_seat, torch, totem\_pole, tow\_truck, toyshop, tractor, trailer\_truck, tray, trench\_coat, tricycle, trimaran, tripod, triumphal\_arch, trolleybus, trombone, tub, turnstile, typewriter\_keyboard, umbrella, unicycle, upright, vacuum, vase, vault, velvet, vending\_machine, vestment, viaduct, violin, volleyball, waffle\_iron, wall\_clock, wallet, wardrobe, warplane, washbasin, washer, water\_bottle, water\_jug, water\_tower, whiskey\_jug, whistle, wig, window\_screen, window\_shade, Windsor\_tie, wine\_bottle, wing, wok, wooden\_spoon, wool, worm\_fence, wreck, yawl, yurt, web\_site, comic\_book, crossword\_puzzle, street\_sign, traffic\_light, book\_jacket, menu, plate, guacamole, consomme, hot\_pot, trifle, ice\_cream, ice\_lolly, French\_loaf, bagel, pretzel, cheeseburger, hotdog, mashed\_potato, head\_cabbage, broccoli, cauliflower, zucchini, spaghetti\_squash, acorn\_squash, butternut\_squash, cucumber, artichoke, bell\_pepper, cardoon, mushroom, Granny\_Smith, strawberry, orange, lemon, fig, pineapple, banana, jackfruit, custard\_apple, pomegranate, hay, carbonara, chocolate\_sauce, dough, meat\_loaf, pizza, potpie, burrito, red\_wine, espresso, cup, eggnog, alp, bubble, cliff, coral\_reef, geyser, lakeside, promontory, sandbar, seashore, valley, volcano, ballplayer, groom, scuba\_diver, rapeseed, daisy, yellow\_lady's\_slipper, corn, acorn, hip, buckeye, coral\_fungus, agaric, gyromitra, stinkhorn, earthstar, hen-of-the-woods, bolete, ear, toilet\_tissue

## Phase 2: Inference Performance Metrics

|                                    |                     |
|------------------------------------|---------------------|
| Total Images Processed             | 5                   |
| Preprocess Latency                 | 16.36 ms per image  |
| Postprocess Latency                | 108.94 ms per image |
| Peak Memory Usage                  | 246.00 MB           |
| Inference Latency (Without Warmup) | 58.71 ms per image  |
| Inference Latency (With Warmup)    | 0.43 ms per image   |
| Throughput (Without Warmup)        | 5.43 Images/sec     |
| Throughput (With Warmup)           | 7.95 Images/sec     |
| Total Run Time (Without Warmup)    | 920.07 ms           |
| Total Run Time (With Warmup)       | 628.67 ms           |

## Phase 3: Detailed Predictions Summary

| Image File                         | Predicted Class | Confidence |
|------------------------------------|-----------------|------------|
| Cat_November_2010-1a.jpg.webp      | tiger cat       | 39.4%      |
| images.jpeg                        | Saluki          | 79.2%      |
| images1.jpeg                       | tiger           | 75.5%      |
| istockphoto-1189925691-612x612.jpg | passenger car   | 73.7%      |
| tigermain.jpg                      | tiger           | 94.1%      |